An Adaptive Edge-Guided Interface Scheme for 1D Fluid Conservation Laws

Aditya Kaul
Computer Science and Engineering
Aspiring Scholars Directed Research Program
Fremont, USA
0009-0002-6627-5878

Naga Sandeep Chintalapati
Computer Science and Engineering
Aspiring Scholars Directed Research Program
Fremont, USA
0009-0000-6888-9246

Vunal Jinasundera
Computer Science and Engineering
Aspiring Scholars Directed Research Program
Fremont, USA
0009-0007-0626-1841

***Abstract*—This work presents the Adaptive Edge-Guided Interface Scheme (`AEGIS`), a high-order shock-capturing scheme for the 1D Euler equations. `AEGIS` adaptively blends fifth-order and low-order numerical fluxes based on local flow smoothness. The proposed edge-guided blending method combines a modified Jameson–Schmidt–Turkel discontinuity sensor with a Weibull mapping. This approach uses both pressure and density indicators to detect discontinuities, while the mapping converts the resulting interface indicators into flux weights. `AEGIS` is evaluated on the Sod, Lax, and Shu–Osher shock-tube problems and a smooth Gaussian density-pulse advection problem. The L_1 and L_2 errors of `AEGIS` are compared with those of a first-order Lax–Friedrichs solver and a component-wise finite-difference implementation of the fifth-order Jiang–Shu weighted essentially non-oscillatory scheme (WENO5-JS) for the tested problems. Across the shock-tube problems, `AEGIS` reduced density-profile L_1 errors by 15.71–31.59% and L_2 errors by 0.15–25.71% relative to WENO5-JS. For the smooth advection problem, `AEGIS` exhibited approximately fifth-order convergence. The results demonstrate that `AEGIS` preserves high-order accuracy in smooth regions while robustly capturing discontinuities.**

***Index Terms*—computational fluid dynamics, high-order methods, finite-volume methods, shock-capturing, 1D Euler equations.**

I. Introduction

Computational fluid dynamics (CFD) plays an essential role in the simulation and design of engineering systems by reducing the cost of building real-life physical prototypes. However, numerical CFD schemes face a fundamental trade-off between high-order accuracy and robust stability in problems with strong discontinuities [1].

High-order methods are often preferred for modeling smooth flow due to their ability to more accurately reconstruct interface flux values by using more information about a cell's surroundings. However, these approaches often face the issue of creating spurious oscillations in the presence of discontinuities, thereby degrading solution accuracy [1]. Conversely, low-order methods are generally more dissipative, resulting in solutions which smear important flow features, thus reducing accuracy [1]. At the same time, these methods are far more robust across discontinuities [1]. While the accuracy issue can be mitigated through mesh refinement, the downside of this approach is significantly higher computational cost [1].

979-8-3195-3759-1/26/$31.00 ©2026 IEEE

The nature of these problems has motivated researchers to spend decades devising numerical schemes to maximize solution accuracy while maintaining numerical stability across shocks. One strategy is to use high-order, low-dissipation methods in regions of smooth flow and gradually increase numerical dissipation near detected discontinuities [2], [3]. Many modern approaches employ a related adaptive strategy, such as weighted essentially non-oscillatory (WENO) schemes which use nonlinear weighted combinations of multiple candidate reconstructions from various substencils [3]. The weights are usually calculated from local smoothness indicators, so that smoother substencils have their reconstructions more heavily weighted [3].

This work presents an alternative 1D shock-capturing numerical scheme: the Adaptive Edge-Guided Interface Scheme (`AEGIS`). `AEGIS` combines a modified Jameson–Schmidt–Turkel (JST) discontinuity sensor [2] with a Weibull mapping to adaptively blend higher- and lower-order numerical fluxes at cell edges (interfaces) according to local flow smoothness. The governing Euler equations for 1D fluid flow are first introduced, followed by the numerical formulation of `AEGIS`. The scheme is then evaluated using standard 1D benchmark problems, and its results are compared to those of traditional CFD schemes. Finally, the accuracy of `AEGIS` relative to the tested baselines is discussed, and directions for future work are outlined.

II. Methods

`AEGIS` solves the 1D Euler equations in conservative form. These hyperbolic conservation laws govern fluid motion by describing the conservation of mass (represented by density, ρ), momentum (ρu) and total energy (E) for a compressible fluid. These equations (1) also require that fluid discontinuities satisfy the Rankine–Hugoniot conditions [1]. The conservative-variable vector $U = [\rho, \rho u, E]^T$ represents the conserved quantities per unit volume and has the corresponding flux vector $F(U) = [\rho u, \rho u^2 + p, u(E+p)]^T$ [1]. The finite-volume update is performed using conservative variables so that mass,

momentum, and energy remain discretely conserved and discontinuities propagate consistently with the Rankine–Hugoniot conditions. Interface reconstruction, however, is performed using the primitive-variable vector $W = [\rho, u, p]^T$, where u represents velocity and p represents pressure, permitting the direct positivity checks on density and pressure described later.

$$\frac{\partial U}{\partial t} = -\frac{\partial F(U)}{\partial x} \tag{1}$$

A. Numerical Flux Calculation

To solve for the flux numerically, AEGIS utilizes an approach which blends together numerical fluxes of three different orders of accuracy: first, second, and fifth. The first-order flux uses constant cell states combined with a Rusanov flux solver [1], given by (2), where $F_{i+1/2}$ is the flux at the cell interface and S^+ is the local maximum wave speed. U_L and U_R represent the left and right states of U at interface $x_{i+1/2}$, respectively, and thus $F_L = F(U_L)$ and $F_R = F(U_R)$.

$$F_{i+1/2} = \frac{1}{2}(F_L + F_R) - \frac{1}{2}S^+(U_R - U_L) \tag{2}$$

For the second-order method, AEGIS uses the Monotonic Upstream-centered Scheme for Conservation Laws (MUSCL) with a Monotonized Central (MC) slope limiter to reconstruct the left and right interface states for each of the primitive quantities [4]. The fifth-order primitive-variable reconstruction is given by Equations (3) and (4) [5].

$$W_{L,i} = \frac{2W_{i-2} - 13W_{i-1} + 47W_i + 27W_{i+1} - 3W_{i+2}}{60} \tag{3}$$

$$W_{R,i} = \frac{2W_{i+2} - 13W_{i+1} + 47W_i + 27W_{i-1} - 3W_{i-2}}{60} \tag{4}$$

For the reconstruction centered on cell i, $W_{L,i}$ denotes the left state reconstructed at $x_{i+1/2}$, while $W_{R,i}$ denotes the right state reconstructed at $x_{i-1/2}$. If the reconstructed pressure or density falls below the positivity floor of 10^{-12}, that quantity is replaced with its corresponding cell-centered value [6]. AEGIS then converts these primitive quantities to their conservative counterparts. At $x_{i+1/2}$, $U_{L,i}$ and $U_{R,i+1}$ are paired to form the left and right interface states U_L and U_R, respectively. These states are then given to the Harten–Lax–van Leer–Contact (HLLC) solver to calculate numerical fluxes at the cell interfaces, as given by Equation (5) [1].

$$F_{i+1/2} = \begin{cases} F_L & 0 \le S_L \\ F_{*L} & S_L \le 0 \le S_* \\ F_{*R} & S_* \le 0 \le S_R \\ F_R & 0 \ge S_R \end{cases} \tag{5}$$

Here, F_L and F_R are the physical fluxes evaluated from U_L and U_R, respectively. F_{*L} and F_{*R} are computed through

$$F_{*K} = F_K + S_K(U_{*K} - U_K), \qquad K \in \{L, R\} \tag{6}$$

where U_{*K} denotes the corresponding intermediate state obtained from the standard HLLC Rankine–Hugoniot relations [1]. $S_L = \min(u_L - c_L,\ u_R - c_R)$ and $S_R = \max(u_L + c_L,\ u_R + c_R)$ estimate the speeds of the fastest local left- and right-moving waves, respectively, where $c_K = \sqrt{\gamma p_K / \rho_K}$ is the local sound speed [1]. A value of $\gamma = 1.4$ is used for the ratio of specific heats. The contact wave speed S_* is then estimated by Equation (7).

$$S_* = \frac{p_R - p_L + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)} \tag{7}$$

After calculating the three candidate numerical fluxes, AEGIS defines the fifth-order flux as the high-order flux (HOF) and utilizes a 3:1 blend of the second- and first-order fluxes for its lower-order flux (LOF). Through numerical experimentation, it was found that using a 75% second-order flux and a 25% first-order flux allowed for accurate and robust shock-capturing.

B. The Blending Scheme

To detect local discontinuities within the fluid, AEGIS utilizes a modified JST-type discontinuity sensor [2], extended from a three-point stencil to a five-point stencil. This extension uses the normalized fourth undivided difference in place of the normalized second undivided difference formula that the original JST sensor uses. This aligns the sensor with the reconstruction's stencil, allowing discontinuities within that stencil to trigger additional numerical dissipation.

The sensor (8) is applied separately to pressure and density ($q \in \{p, \rho\}$) on a five-point stencil centered at i and returns a value ($\nu_{q,i}$) between 0 and 1. Values near zero indicate locally smooth flow, whereas larger indicator values indicate stronger local discontinuities. While the pressure sensor $\nu_{p,i}$ detects shocks, the density sensor $\nu_{\rho,i}$ detects sharp changes in density, including contact discontinuities across which pressure remains continuous [1].

$$\nu_{q,i} = \left|\frac{q_{i-2} - 4q_{i-1} + 6q_i - 4q_{i+1} + q_{i+2}}{q_{i-2} + 4q_{i-1} + 6q_i + 4q_{i+1} + q_{i+2}}\right| \tag{8}$$

Given that pressure and density remain positive, the denominator of (8) will always be greater than or equal to the absolute value of the numerator. Therefore, $0 \le \nu_{q,i} \le 1$, preserving the bounded nature of the original JST sensor [2].

After calculating pressure and density discontinuity indicator values for each cell, AEGIS combines them into a single cell-centered discontinuity indicator (ν_i) through

$$\nu_i = \max(\nu_{p,i}, \nu_{\rho,i}) \tag{9}$$

To obtain an interface-centered indicator, AEGIS then calculates $\nu_{i+1/2}$ as the maximum of indicator values in cells i and $i+1$, as given by Equation (10).

$$\nu_{i+1/2} = \max(\nu_i, \nu_{i+1}) \tag{10}$$

The interface indicator is then mapped to the weight of the LOF ($\omega_{i+1/2}$) using the Weibull cumulative distribution function given by Equation (11).

$$\omega_{i+1/2} = 1 - e^{-(\nu_{i+1/2}/\lambda)^k} \tag{11}$$

The shape and scale parameters are set to $k = 2$ and $\lambda = 0.02$, respectively, and are held fixed across all problems. These values were tuned empirically through preliminary trials to increase the LOF (F^{LO}) weight rapidly with discontinuity strength, balancing accuracy and stability across the tested problems. This mapping guarantees $0 \leq \omega_{i+1/2} \leq 1$, with values near 0 favoring the HOF (F^{HO}) in smooth regions and values approaching 1 favoring the LOF near discontinuities. While $k = 2$ is selected to provide a smooth, nonlinear increase in the LOF weight, the choice of $\lambda = 0.02$ is evaluated later in Section IV-E.

Finally, `AEGIS` calculates the blended numerical flux $F_{i+1/2}$ at each interface used to update the interior (non-boundary) cells according to Equation (12).

$$F_{i+1/2} = \omega_{i+1/2} F^{\mathrm{LO}}_{i+1/2} + (1 - \omega_{i+1/2}) F^{\mathrm{HO}}_{i+1/2} \tag{12}$$

C. Right Hand Side Calculation

After calculating $F_{i+1/2}$ for each cell interface, `AEGIS` calculates the right-hand side (RHS) of Equation (1) [1] for each interior cell as

$$\mathrm{RHS}_i = -\frac{F_{i+1/2} - F_{i-1/2}}{dx}, \tag{13}$$

where dx is the cell length.

D. Time Integration

To advance the solution in time, `AEGIS` employs a fourth-order Runge–Kutta (RK4) time-stepping approach [1]:

$$k_1 = \mathrm{RHS}(U^j) \tag{14a}$$

$$k_2 = \mathrm{RHS}(U^j + \frac{dt}{2} \cdot k_1) \tag{14b}$$

$$k_3 = \mathrm{RHS}(U^j + \frac{dt}{2} \cdot k_2) \tag{14c}$$

$$k_4 = \mathrm{RHS}(U^j + dt \cdot k_3) \tag{14d}$$

$$U^{j+1} = U^j + \frac{dt}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{14e}$$

Here, j is the current time-step index, $\mathrm{RHS}(U^j)$ denotes the vector of RHS values (RHS_i) evaluated from the vector U^j, and dt is the size of the time step. For each time step, the size of dt for the next step is calculated from the current solution using the Courant–Friedrichs–Lewy (CFL) condition [1]

$$dt = \frac{C \cdot dx}{\max_i(|u_i| + c_i) + \epsilon}, \tag{15}$$

where C is the CFL number used for the simulation and $\epsilon = 10^{-12}$ to prevent division by zero.

TABLE I
DENSITY-PROFILE L_1 AND L_2 ERRORS FOR THE LAX, SHU–OSHER, AND SOD SHOCK-TUBE PROBLEMS.

Problem	**Method**	L_1 **Error**	L_2 **Error**
Lax	FO-LF	0.088 797	0.158 258
	WENO5-JS	0.009 026	0.040 616
	AEGIS	0.007 579	0.040 554
Shu–Osher	FO-LF	0.201 766	0.352 309
	WENO5-JS	0.078 681	0.179 534
	AEGIS	0.053 825	0.133 374
Sod	FO-LF	0.031 085	0.041 717
	WENO5-JS	0.002 646	0.008 024
	AEGIS	0.002 230	0.007 682

III. Problem Setup & Results

A. Compared Methods

`AEGIS` is compared with two established CFD schemes. The first method is the component-wise finite-difference implementation of the fifth-order Jiang–Shu weighted essentially non-oscillatory scheme (WENO5-JS) [3]. For consistency with `AEGIS`, WENO5-JS is advanced in time using RK4. The second is the standard first-order Lax–Friedrichs scheme with first-order time-stepping (FO-LF) [1].

All three schemes are tested on three shock-tube problems and a smooth-advection problem to evaluate the comparative performance of `AEGIS` in both discontinuous and smooth flow.

B. Problem Setup

All problems were run with a CFL number of $C = 0.4$. Here, all reported resolutions include six boundary cells. Each scheme was run at a resolution of $N = 256$ cells for the shock-tube problems. For the convergence test, accuracy was evaluated at $N = 32, 64, 128, 256, 512,$ and 1024. While the reference solutions for the shock-tube problems were computed using WENO5-JS at a resolution of $N = 4096$, the reference solution for the convergence test was computed using WENO5-JS at a resolution of $N = 10240$. These resolutions ensured that the reference solutions were significantly finer than the solutions being evaluated. Zero-gradient outflow boundary conditions were applied to all problems.

The initial conditions for all tested problems are described in Table II. Initial primitive variable vector states are listed as (ρ, u, p). For the shock-tube problems, left states apply for $x < x_0$ and right states for $x \geq x_0$, where x_0 represents the discontinuity position within the domain. Domain lengths and final times for each problem are specified below. All problems use uniform spatial grids over $x \in [0, L]$.

1) The Lax Shock-Tube: The first test used for comparison is the Lax shock-tube problem. The domain length is set to $L = 1.0$, and the final results are computed at $t = 0.14$.

2) The Shu–Osher Shock-Tube: The second test used to validate `AEGIS` is a normalized form of the Shu–Osher shock-tube problem. The domain length is again set to $L = 1.0$, and the final results are computed at $t = 0.18$.

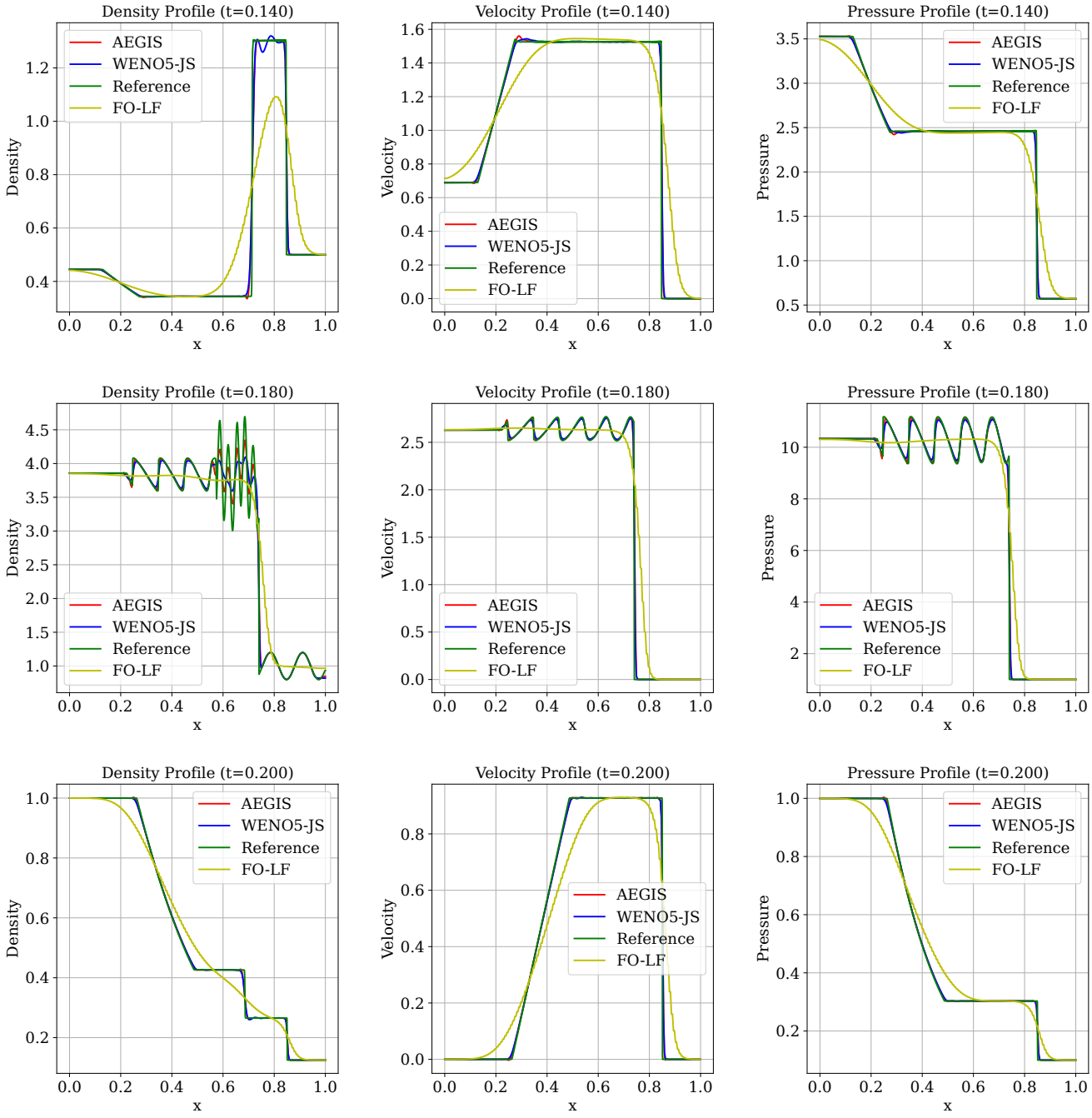


Fig. 1. Density, velocity, and pressure distributions for the Lax (top), Shu–Osher (middle), and Sod (bottom) shock-tube problems.

TABLE II
INITIAL CONDITIONS FOR THE TEST PROBLEMS. LEFT STATES APPLY FOR $x < x_0$ AND RIGHT STATES FOR $x \geq x_0$.

Problem	x_0	Left state (ρ, u, p)	Right state (ρ, u, p)
Lax	0.5	$(0.445,\ 0.689,\ 3.528)$	$(0.5,\ 0,\ 0.571)$
Shu–Osher	0.1	$(27/7,\ 2.629369,\ 31/3)$	$(1 + 0.2\sin(50x),\ 0,\ 1)$
Sod	0.5	$(1,\ 0,\ 1)$	$(0.125,\ 0,\ 0.1)$
Smooth advection	—	$\left(1 + 0.1\exp\left[-\frac{(x-1)^2}{2(0.1)^2}\right],\ 1,\ 1\right)$	

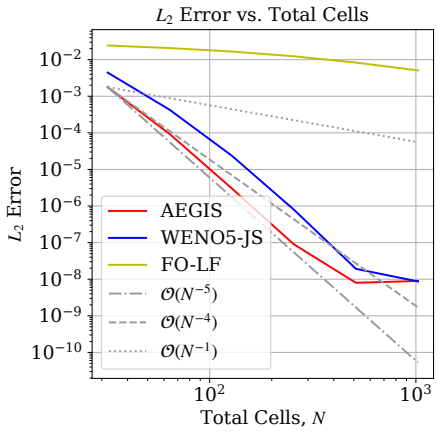


Fig. 2. L_2 convergence plot for the smooth advection problem.

3) The Sod Shock-Tube: The third test is the classic Sod shock-tube. Like the previous two problems, the domain length is set to $L = 1.0$, and the final results are computed at $t = 0.2$.

4) 1D Smooth Advection: The final test is the advection of a Gaussian density pulse. The domain length is set to $L = 2.0$, and the final results are computed at $t = 0.5$ to limit the computational cost of generating the reference solution.

C. Problem Outputs

Table I gives the density-profile L_1 and L_2 errors for each of the shock-tube problems, rounded to six decimal places. The reported L_1 and L_2 errors are calculated as the mean absolute error and root-mean-square error, respectively. Figure 1 gives the density, velocity, and pressure profiles for the Lax (top panels), Shu–Osher (middle panels), and Sod (bottom panels) shock-tube problems.

Figure 2 shows the density-profile L_2 errors versus grid resolution for the smooth 1D advection problem. First-, fourth-, and fifth-order reference lines, corresponding to $\mathcal{O}(N^{-1})$, $\mathcal{O}(N^{-4})$, and $\mathcal{O}(N^{-5})$, respectively, are anchored to the lowest-resolution AEGIS error and included for comparison.

All errors were calculated over interior cells only. The six boundary cells, three at each end of the domain, were excluded to ensure that they did not directly enter the numerical scheme error calculations.

IV. Discussion

A. The Lax Shock-Tube

Regarding the Lax problem, the pressure jump is larger than in Sod. However, AEGIS remains stable and shows only small deviations near the sharp transitions, as seen in the top panels of Figure 1. While remaining stable, AEGIS maintains its accuracy, closely tracking the reference solution for all three profiles, demonstrating its ability to correctly model even larger discontinuities. This shock-capturing behavior is reflected in the error measurements. Compared to WENO5-JS, AEGIS reduces the density-profile L_1 error by 16.03% and produces a comparable L_2 error (0.15% lower).

B. The Shu–Osher Shock-Tube

In the density profile for the Shu–Osher problem (middle-left panel of Figure 1), FO-LF is too dissipative, smearing out the high-frequency density oscillations behind the shock at $t = 0.18$. WENO5-JS appears to be more effective at capturing these features, but it underpredicts the post-shock oscillation amplitudes. In comparison, AEGIS more closely reproduces the interaction between the discontinuity and smooth density waves, which is consistent with how the discontinuity sensor assigns greater weight to the LOF near discontinuities in either pressure or density. Accordingly, AEGIS becomes LOF-dominant only at interfaces where the sensor indicates stronger discontinuities, while maintaining HOF dominance in smoother regions. The density-profile error measurements reflect this as well, with AEGIS having 31.59% lower L_1 error and 25.71% lower L_2 error, as compared to WENO5-JS.

C. The Sod Shock-Tube

For the Sod problem, FO-LF once again produces substantial smearing near discontinuities because of its high numerical dissipation. However, WENO5-JS and AEGIS capture these wave structures more accurately, as seen in the bottom panels of Figure 1. The lower error of AEGIS relative to WENO5-JS is consistent with its discontinuity sensor remaining near zero in smooth regions, thereby favoring the high-order flux. Near pressure or density discontinuities, AEGIS rapidly increases the LOF weight. This allows AEGIS to accurately capture both the smooth rarefaction and the discontinuities within the density profile, while the velocity and pressure profiles also closely follow the reference solution. As a result, compared to WENO5-JS, AEGIS reduces the L_1 error by 15.71% and the L_2 error by 4.26%.

D. Convergence Study

In Figure 2, AEGIS demonstrates its ability to maintain high-order accuracy for smooth flows such as the 1D smooth advection problem. Before plateauing, the L_2 error of AEGIS decreases more rapidly than the first- and fourth-order reference lines and exhibits a slope comparable to that of the fifth-order line. Using the pre-plateau data, the observed convergence order for AEGIS is approximately 4.6. The shortfall from a full fifth-order convergence rate may partly reflect the fourth-order RK4 time-stepper, which can limit the convergence order of AEGIS as a whole. Even so, this near-fifth-order behavior is consistent with the discontinuity sensor and weight mapping operating as intended, since AEGIS avoids excessive lower-order dissipation in this smooth-flow problem.

Before reaching their numerical error floors, the L_2 error of AEGIS remains below that of WENO5-JS. This indicates that AEGIS can achieve accuracy comparable to WENO5-JS at lower grid resolutions.

The convergence plot also shows the L_2 errors of AEGIS and WENO5-JS plateauing near 10^{-8} at the highest resolutions tested. This indicates that computed errors have reached a numerical error floor that may stem from the limits of numerical precision.

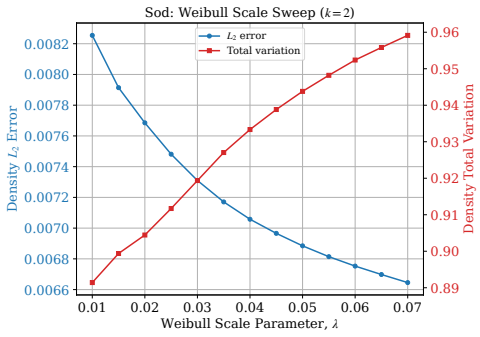


Fig. 3. Density-profile L_2 error and total variation ($\sum_i |\rho_{i+1} - \rho_i|$) for different Weibull scale parameters λ for the Sod shock-tube problem at $t = 0.2$.

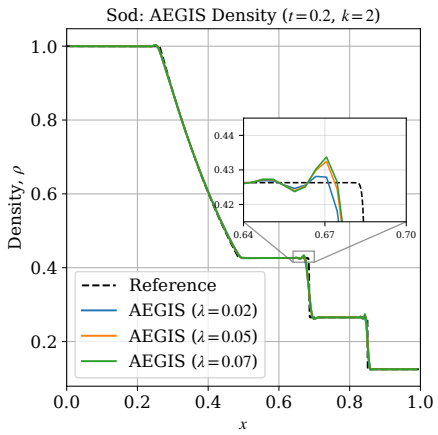


Fig. 4. AEGIS density profiles for different Weibull scale parameters in the Sod shock-tube problem at $t = 0.2$.

E. Weibull Parameter Sensitivity

As mentioned previously, the Weibull shape parameter was set to $k = 2$. k controls the sharpness of the transition from high-order to low-order flux. For $k > 1$, the mapping is relatively insensitive to very small sensor values, so weak variations in smooth regions do not immediately add dissipation. As ν approaches λ, the weight then rises more rapidly. With $k = 2$, the mapping exhibits a quadratic response near $\nu = 0$, which suppresses unnecessary low-order activation in smooth regions while still giving a fairly sharp transition near detected discontinuities.

With regard to λ, values from 0.010 to 0.070 were tested with $k = 2$ held fixed for the Sod shock-tube problem. Figure 3 shows that density-profile L_2 errors decrease as the scale parameter increases. However, at these higher values, AEGIS was observed to generate slight numerical oscillations near discontinuities, as seen in Figures 3 and 4. This may be because these larger parameter values were slower to increase the LOF weight around discontinuities, thereby reducing numerical dissipation and allowing oscillations to develop. As a result, $\lambda = 0.02$ was found to be a practical compromise between maintaining low error and limiting nonphysical oscillations, and was thus selected for use in mapping discontinuity indicator values to the LOF weights.

V. Conclusion

Resolving complex flows in CFD continues to require balancing accuracy with stability near discontinuities. Across the shock-tube problems tested, AEGIS achieved lower density-profile L_1 and L_2 errors than both FO-LF and WENO5-JS while maintaining stable shock capturing. These improvements may reflect the combined effects of the modified JST-type discontinuity sensor, flux-blending scheme, and primitive-variable reconstruction. AEGIS also maintained approximately fifth-order convergence for smooth 1D advection. Together, these results demonstrate that the method preserves high-order accuracy in smooth regions while accurately and stably capturing discontinuities.

These results are limited, however, given that testing has only been performed on four 1D Euler benchmarks. Additionally, this means that any multidimensional application of AEGIS requires separate validation. Future work on AEGIS will investigate alternative discontinuity sensors and functions to map indicators to flux weights, with the goal of further reducing numerical dissipation while still preserving accuracy and stability. Additional work will also include researching ways to reduce computational costs, extending the scheme to model the 2D Euler equations, and eventually generalizing AEGIS to magnetohydrodynamic systems.

The source code and additional numerical results and plots are available in the AEGIS repository at:
https://github.com/ASDRPScholars/AEGIS.

Acknowledgment

The authors acknowledge Christopher DeGrendele for the invaluable mentoring and insights that made this work possible. The authors also thank ASDRP for the opportunity to conduct this research.

References

[1] E. F. Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics: A Practical Introduction*. Berlin, Germany: Springer, 3rd ed., 2009.

[2] J. Duan, W. Barsukow, and C. Klingenberg, "Active flux methods for hyperbolic conservation laws—flux vector splitting and bound-preservation," *SIAM Journal on Scientific Computing*, vol. 47, no. 2, pp. A811–A837, 2025.

[3] G.-S. Jiang and C.-W. Shu, "Efficient implementation of weighted ENO schemes," *Journal of Computational Physics*, vol. 126, no. 1, pp. 202–228, 1996.

[4] J. Holman, "Comparison of different limiters in MUSCL reconstruction for the numerical solution of compressible flow," in *Topical Problems of Fluid Mechanics 2025* (D. Šimurda and T. Bodnár, eds.), (Prague, Czech Republic), pp. 115–122, 2025.

[5] J. Mu, C. Zhuo, Q. Zhang, S. Liu, and C. Zhong, "An efficient high-order gas-kinetic scheme with hybrid WENO-AO method for the Euler and Navier–Stokes solutions," *Communications in Computational Physics*, vol. 35, no. 3, pp. 662–692, 2024.

[6] P. S. Verma, J.-M. Teissier, O. Henze, and W.-C. Müller, "Fourth-order accurate finite-volume CWENO scheme for astrophysical MHD problems," *Monthly Notices of the Royal Astronomical Society*, vol. 482, pp. 416–437, 01 2019.